

[phf] Power-High-Feed

High feed milling for roughing applications on high-powered machines.



Tools. Simply. Different.

Product overview

The [phf] Power High Feed stands for the highest machining performance for Contour- and Face Milling. Compared to conventional machining it's possible to achieve a considerably higher table feed with this tool type. There are two insert sizes available for different machine stability and applications.

SK40 and HSK63, in machines without ceramic bearing: PHF09 (9mm)
SK50 > HSK63 in high power machines also in spindle-axis: PHF14 (14mm)

Technical characteristics

The four indexable insert with a positive soft cutting geometry is available in two carbide grades. The entering angle of the tool can give extreme high feed rates at small axial cutting depths. In addition to this the main part of the cutting force is directed axially towards the machine spindle. This allows stable machining up to 4xD. To utilize this tool a stable, high-performance machine that can handle a high feed force in the **spindle direction** is needed.

Applications

Main application: Contour roughing of 3D surfaces in soft and tempered steel up to 45HRC.

Secondary application: Face milling, plunge milling and helical interpolation.

Material	Grade	Cutting speed v_c (m/min)	PHF09 a_p (mm)	PHF09 f_z (mm)	PHF14 a_p (mm)	PHF14 f_z (mm)
unalloyed steel	P25U/K15	180-240	0,8-1,2	0,8-1,5	1,2-2,0	0,8-2,0
alloyed steel <1100N	P25U/K15	120-180	0,8-1,2	0,8-1,5	1,2-1,8	0,8-2,0
grey cast	P25U/K15	160-250	0,8-1,2	0,8-1,5	1,2-2,0	0,8-2,0
pre hardened steel <45HRC	K15	70-110	0,4-0,8	0,5-1,0	0,4-1,0	0,5-1,5
Titanium	K15	35-45	0,8-1,2	0,5-1,2	1,0-1,5	0,5-1,5
ChrNi	K15	160-220	0,8-1,2	0,7-1,5	1,0-1,8	0,5-1,5

Please contact your AURA® sales contact in advance for your first need in this application.

Bauformen



Order-Code	Size	D1 (mm)	D2 (mm)	R (mm)	Z	X1	IKZ	max. Ramp	min. hole diameter (mm)
21670	PHF09	42	29	2	4	M16	yes	2°	68
21671	PHF09	52	40	2	5	AST22	yes	2°	88
21672	PHF09	66	48	2	5	AST27	yes	1,5°	116
21673	PHF14	66	48	3	5	AST27	yes	2,5°	107
21674	PHF14	80	60	3	6	AST27	yes	2°	135
21675	PHF14	100	70	3	6	AST32	yes	1,5°	175

Order-Code	Insert	Size	Grade	Geometry	f_z (mm) [min-max]	a_p (mm) [max]
21745	PHF	09	K15	positive	0,4-1,7	1,2
22295	PHF	09	P25U	positive	0,4-1,7	1,2
21746	PHF	14	K15	positive	0,4-2,5	2,0
22296	PHF	14	P25U	positive	0,4-2,5	2,0

Advantages

- The most productive solution when highest metal removal rate is needed.
- Very silent cutting conditions, also at 90° surface.
- High-strength HQC cutter bodies with master quality.
- Grades for a wide application area K15 and P25U.
- Can be machined with a tool length up to 4xD.
- Two insert sizes PHF09, PHF14.
- Inserts with positive geometry.
- Internal coolant.
- Suitable for interrupted cuts and 90° walls.

ExpertTipp

Use max. a_e to get a more quietness and stable conditions. If encountering a material with less machinability we recommend to reduce the radial engagement a_e . Ramping and helix is recommended at an angle of 1 - 1,5°. When entering sharp corners the feedrate can be reduced to about 50%.



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- developed
- + produced
- in Germany